TESTING UNIVERSAL CLOSEDNESS ON NORMAL CURVES PUNCTURED AT ONE POINT (NAGATA BASE)

THE STACKS PROJECT AUTHORS

Editorial note. Verbatim author omission in the counted span: 'Then we claim that the restriction f' : X' -> Y' of f viewed as a morphism of schemes over S' has property (3) also. We omit the details.' The author flags the paragraph 'we suggest skipping'; it is the mechanical locality of condition (3), while the load-bearing Chevalley/Nagata argument is fully worked. Flagged needs_human_spotcheck.

Allowed standard prerequisites (editorial).

- Nagata normalization finiteness. Morphisms, Lemma morphisms-lemma-nagata-normalization-finite-general

Source reference. Original reference 00PD: sources/stacks-chow/tags.txt, source lines 927–927. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GWT: sources/stacks-chow/tags.txt, source lines 21083–21083. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05JY: sources/stacks-chow/tags.txt, source lines 7182–7182. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02J3: sources/stacks-chow/tags.txt, source lines 3322–3322. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 054J: sources/stacks-chow/tags.txt, source lines 6642–6642. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 054I: sources/stacks-chow/tags.txt, source lines 6641–6641. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02J6: sources/stacks-chow/tags.txt, source lines 3325–3325. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 005X: sources/stacks-chow/tags.txt, source lines 225–225. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01TF: sources/stacks-chow/tags.txt, source lines 2372–2372. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original reference 01J4: sources/stacks-chow/tags.txt, source lines 2001–2001. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BAK: sources/stacks-chow/tags.txt, source lines 14213–14213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 035A: sources/stacks-chow/tags.txt, source lines 4121–4121. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0AVK: sources/stacks-chow/tags.txt, source lines 13688–13688. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BX8: sources/stacks-chow/tags.txt, source lines 14971–14971. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a Nagata scheme (and in particular locally Noetherian). Let $f : X \to Y$ be a quasi-compact morphism of schemes locally of finite type over S. The following are equivalent*

(1) *f is universally closed,*
(2) *for every n the morphism $\mathbf{A}^n \times X \to \mathbf{A}^n \times Y$ is closed,*
(3) *for any commutative diagram*

$$\begin{array}{ccc} U & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow f \\ C & \longrightarrow & Y \end{array}$$

of schemes over S such that
(a) *C is a normal integral scheme of finite type over S,*
(b) *$U = C \setminus \{c\}$ for some closed point $c \in C$,*
(c) *$A = \mathcal{O}_{C,c}$ has dimension 1*[1]
then in the commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow f \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

where $K = Frac(A)$ some dotted arrow exists[2] *making the diagram commute.*

Proof. We have seen the equivalence of (1) and (2) and the fact that these imply (3) in Lemma 05JY. Thus it suffices to prove that (3) implies (2). Observe that if condition (3) holds for $f : X \to Y$, then condition (3) holds for $1 \times f : \mathbf{A}^n \times X \to \mathbf{A}^n \times Y$ (see argument in the proof of Lemma 05JY). Hence it suffices to show that (3) implies that f is closed.

Reduction to the case where Y and S are affine; we suggest skipping this paragraph. Let $S' \subset S$ be an affine open and let $Y' \subset Y$ be an affine open mapping into S'. Set $X' = f^{-1}(Y')$. Then we claim that the restriction $f' : X' \to Y'$ of f viewed as

[1] It follows that A is a discrete valuation ring, see Algebra, Lemma 00PD. Moreover, c maps to a finite type point $s \in S$ and A is essentially of finite type over $\mathcal{O}_{S,s}$.

[2] By Lemma 0GWT this is equivalent to asking for the existence of dotted arrow making the first commutative diagram commute.

a morphism of schemes over S' has property (3) also. We omit the details. Now if we can prove that f' is closed for all choices of S' and Y', then it follows that f is closed. This reduces us to the case discussed in the next paragraph.

Assume S and Y affine. Let $Z \subset X$ be a closed subset. We may and do view Z as a reduced closed subscheme of X. We have to show that $E = f(Z)$ is closed. Pick $y \in Y$ a closed point contained in the closure of $f(Z)$. It suffices to show $y \in E$. We assume $y \notin E$ to get a contradiction. The image $s \in S$ of y is a finite type point of S, see Morphisms, Lemma 02J3. Recall that E is constructible (Morphisms, Lemma 054J). Consider the intersection $\mathrm{Spec}(\mathcal{O}_{Y,y}) \cap E$. This is a constructible subset of the spectrum (Morphisms, Lemma 054I) which doesn't contain the closed point. Since the punctured spectrum $\mathrm{Spec}(\mathcal{O}_{Y,y}) \setminus \{y\}$ is Jacobson (Morphisms, Lemma 02J6), we find a closed point $t \in \mathrm{Spec}(\mathcal{O}_{Y,y}) \setminus \{y\}$ with $t \in E$ (see Topology, Lemma 005X). In other words, $t \in E$ is a point of Y which has an immediate specialization $t \rightsquigarrow y$. As $t \in E$ the scheme theoretic fibre Z_t is nonempty. Choose a closed point $x \in Z_t$. In particular we have $[\kappa(x) : \kappa(t)] < \infty$ by the Hilbert Nullstellensatz (Morphisms, Lemma 01TF).

Denote $T = \overline{\{t\}} \subset Y$ the integral closed subscheme whose underlying topological space is as indicated (Schemes, Definition 01J4). Then $t \in T$ is the generic point. Denote $C \to T$ the normalization of T in $\kappa(x)$, see Morphisms, Section 0BAK (more precisely, $C \to T$ is the normalization of T in x where we view $x = \mathrm{Spec}(\kappa(x)) \to T$ as a scheme over T). Since S is a Nagata scheme, so is T (Morphisms, Lemma 035A). Hence we see that $C \to T$ is finite (Morphisms, Lemma 0AVK). As t is in the image we see that $C \to T$ is surjective (because the image is closed and T is the closure of t in Y). Choose a point $c \in C$ mapping to $y \in T$. Since y is a closed point of T we see that c is a closed point of C. Since $\dim(\mathcal{O}_{T,y}) = 1$ we see that $\dim(\mathcal{O}_{C,c}) = 1$ (the dimension is at least 1 as c is not the generic point of C and at most 1 as $C \to T$ is finite). As the function field of C is $\kappa(x)$ and as x is a point of X, we have a Y-rational map from C to X (see for example Morphisms, Lemma 0BX8). Let $C \supset U \to X$ be a representative (in particular U is nonempty). We may assume $c \notin U$ (replace U by $U \setminus \{c\}$). Since c is a closed point of codimension 1 in the integral scheme C we have $C = U \amalg \{c\} \amalg \Sigma$ for some proper closed subset $\Sigma \subset C$. After replacing C by $C \setminus \Sigma$ we have constructed a commutative diagram as in part (3). By the 2nd footnote in the statement of the lemma, the existence of the dotted arrow produces an extension of the rational map to all of C and we get the contradiction because the image of c will be a point of Z mapping to y. $\square$